Marwadi University	Marwadi University Faculty of Engineering and Technology Department of Information and Communication Technology	
Subject: Cloud Computing	**Aim:** Guided Project-2	
Experiment No: 14	**Date:**	**Enrolment No: 92301733047**

Title: Automated CI/CD Containerized Web Deployment

Candidate Name: Raj Tala, E.N: 92301733047

Technology Stack: Docker, AWS (EC2, ECR), GitHub Actions, Nginx

Domain: Cloud DevOps

Github Link: https://github.com/rajpatel10124/SupportiveSailers

1. Project Overview

The objective of this project is to implement a fully automated **Continuous Integration and Continuous Deployment (CI/CD)** pipeline for a containerized static web application. By leveraging modern DevOps tools, the manual effort of building, pushing, and deploying code is eliminated, ensuring high availability and consistency across environments.

2. Technical Architecture

The project follows a "Push-to-Deploy" architecture:

1. **Source Control:** GitHub (Git)
2. **Continuous Integration:** GitHub Actions (Builds Docker images).
3. **Container Registry:** Amazon ECR (Stores private Docker images).
4. **Continuous Deployment:** GitHub Actions via SSH (Automates deployment).
5. **Compute Environment:** AWS EC2
6. **Web Server:** Nginx (Running inside a Docker container).

3. Implementation Details

A. Containerization (Docker)

A custom **Dockerfile** was created using the nginx:alpine base image to keep the footprint small and secure.

- **Base Image:** nginx:alpine
- **Configuration:** Custom HTML files are copied to /usr/share/nginx/html.

- **Networking:** Exposed on Port 80.

B. Infrastructure Setup (AWS)

- **Amazon ECR:** Created a private repository named static-web to host versioned container images.
- **Amazon EC2:** Provisioned a Linux instance with Docker installed and Security Groups configured to allow HTTP (Port 80) and SSH (Port 22) traffic.

C. Automation (GitHub Actions)

The pipeline is defined in .github/workflows/deploy.yml and triggers on every git push.

- **Phase 1 (Build):** Authenticates with AWS, builds the Docker image, and tags it as latest.
- **Phase 2 (Push):** Uploads the image to Amazon ECR.
- **Phase 3 (Deploy):** Connects to EC2 via SSH, pulls the updated image, stops the old container, and starts the new one.

4. Key Features & Benefits

- **Zero Manual Intervention:** Code goes from the developer's laptop to production automatically.
- **Immutability:** Docker ensures that the environment on the developer's machine is identical to the production server.
- **Security:** Uses GitHub Secrets to store sensitive AWS Access Keys and SSH Private Keys.

5. Challenges & Resolutions

- **Issue:** Authorization errors between GitHub and ECR.
- **Resolution:** Implemented aws-actions/configure-aws-credentials and added proper IAM policies (AmazonEC2ContainerRegistryFullAccess) to the user.
- **Issue:** Docker permission denied on EC2.
- **Resolution:** Added the ubuntu user to the docker group on the EC2 instance.

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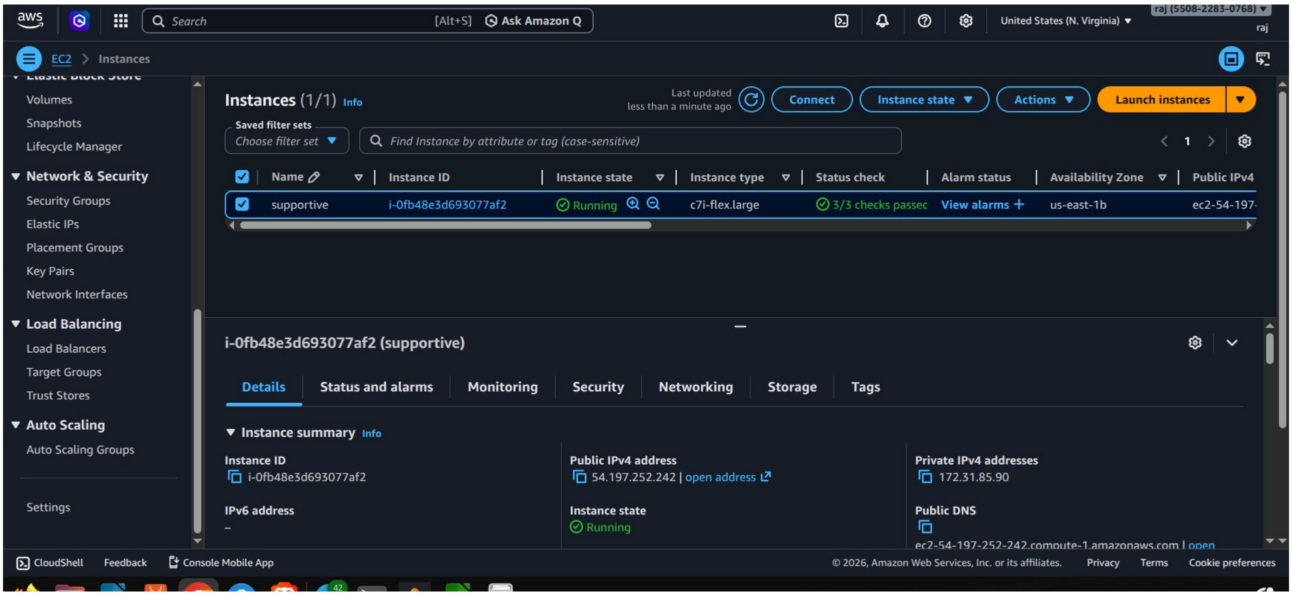
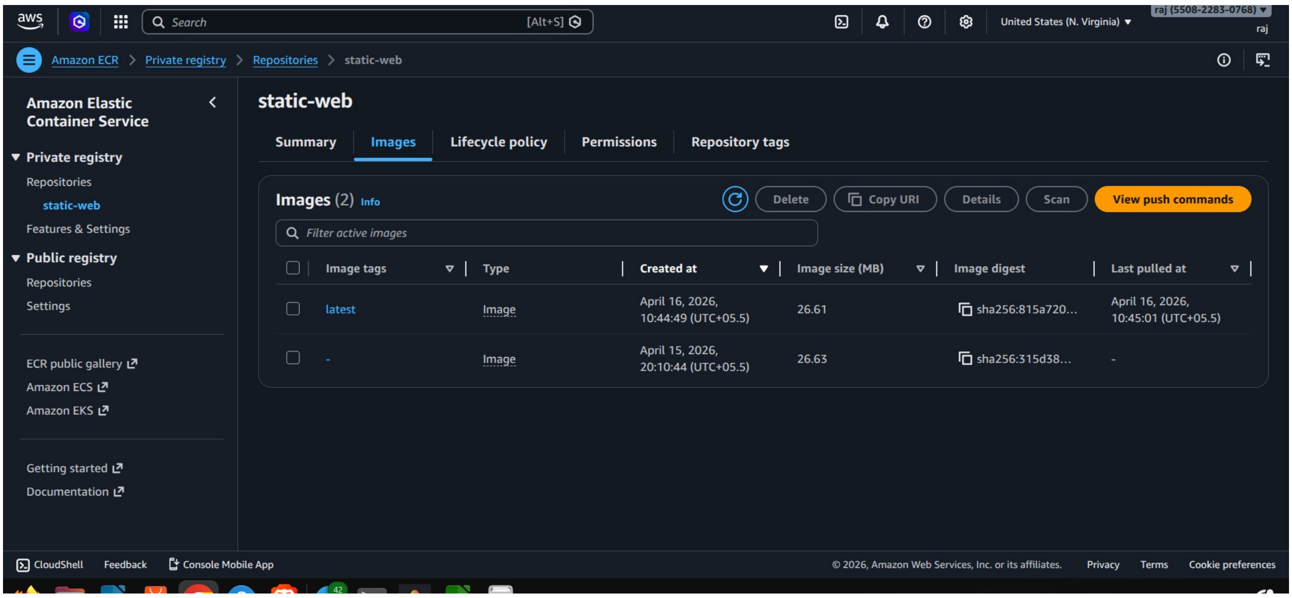
6. Conclusion

This project demonstrates a production-grade DevOps workflow. It showcases the ability to manage cloud infrastructure, write automation scripts, and handle containerized workloads, making it a scalable solution for modern web applications. Website is simple static website but the whole flow is advance level and real time automation flow.





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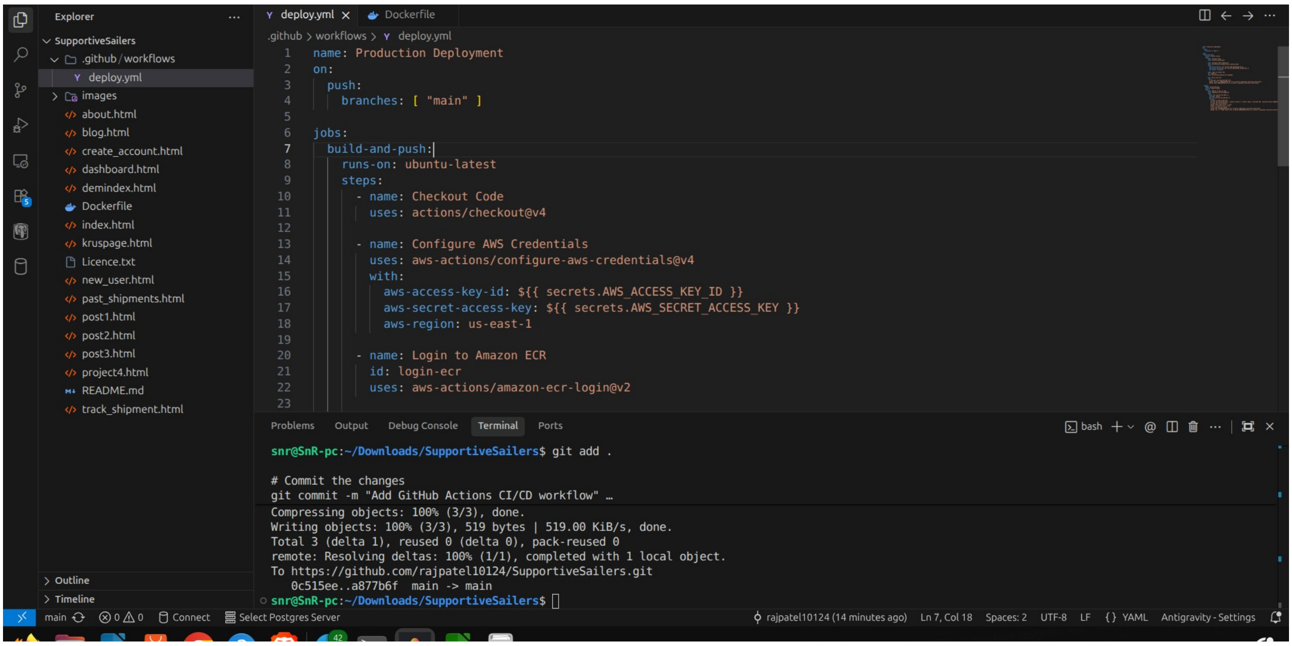

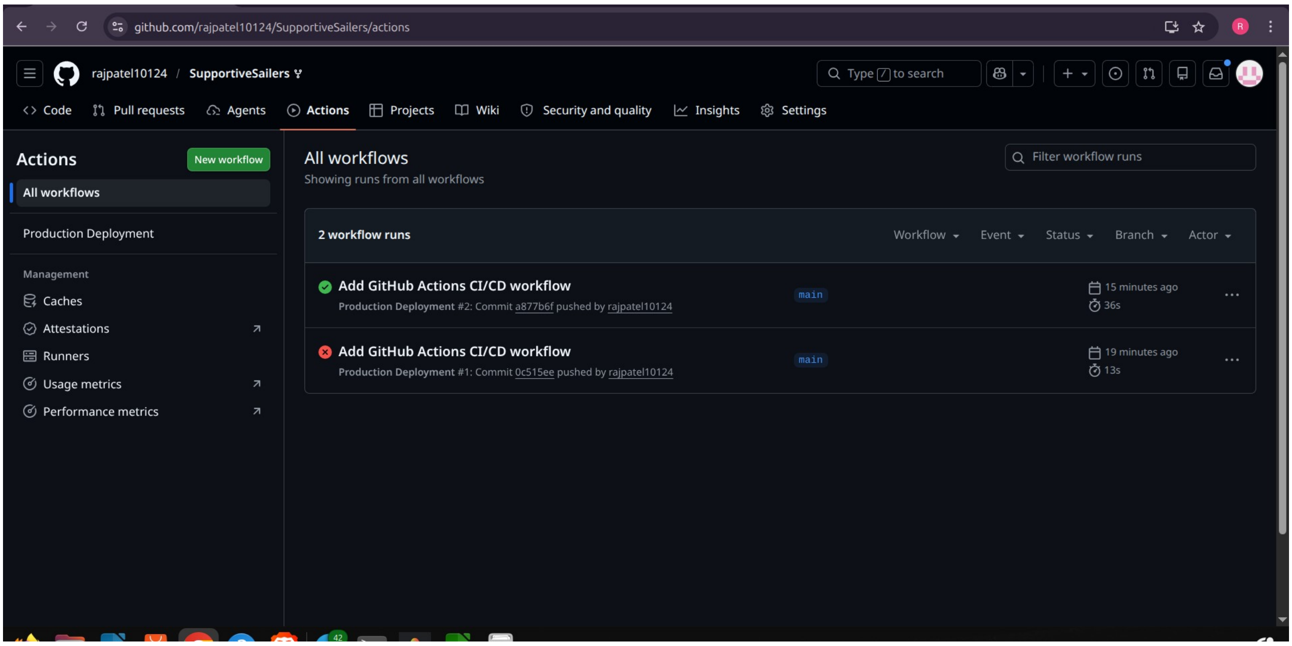

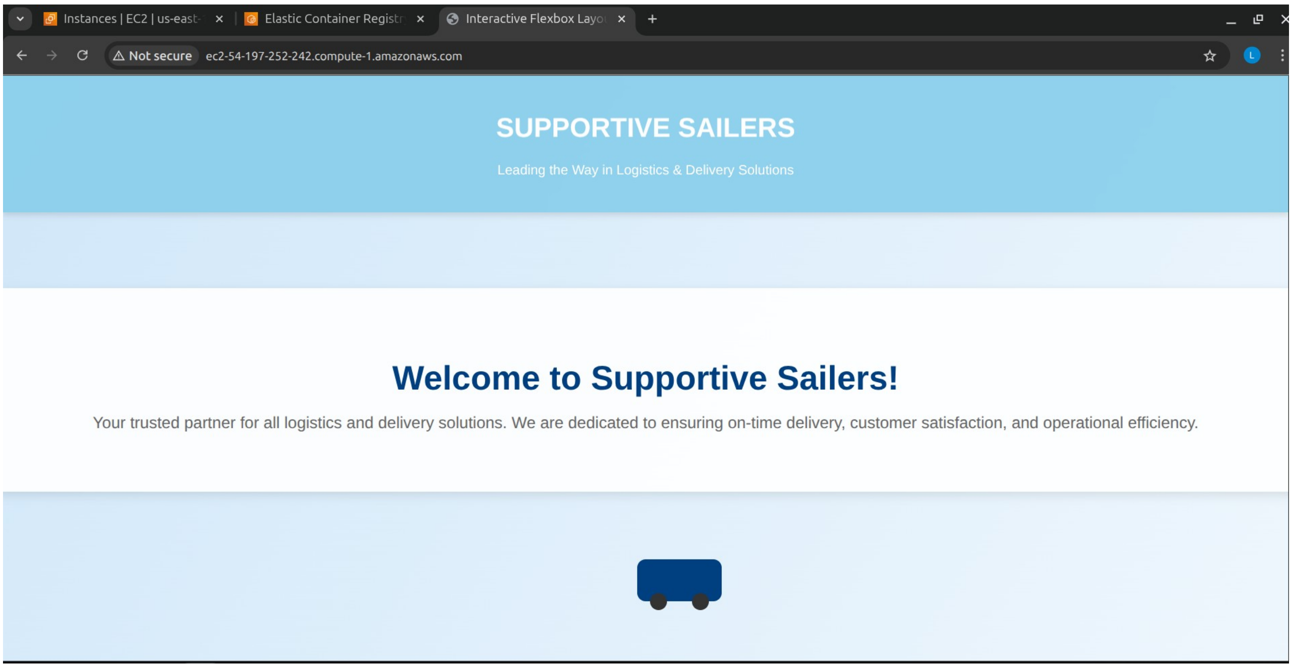

Instances | EC2 | us-east-
Elastic Container Registr
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